DATA SCIENCE • TRAFFIC EARLY WARNING

SafeFlow

Predicting freeway congestion before it starts

Can we warn of congestion early enough to act — without overwhelming operators with false alarms?



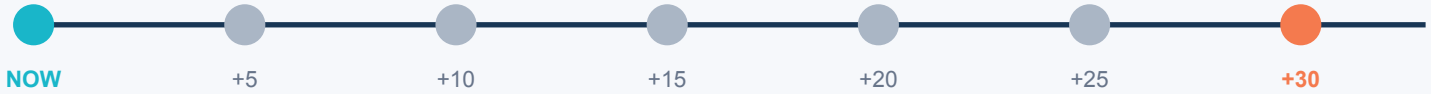
Team SafeFlow | Erdős Institute Data Science Boot Camp | Fall 2026

Current deck uses development evidence only. Final holdout results plug in near the end.

The prediction task: warn before the slowdown arrives

A decision-oriented target turns raw traffic measurements into an early-warning problem.

SENSOR OUTPUT EVERY 5 MINUTES



Inputs at the forecast origin

29 shared features
- speed, flow, occupancy
- recent lags and trends
- spatial / station context
- time-of-day context

Model output

Probability that measured speed falls below 40 mph at any point in the next 30 minutes

Why this framing matters

Predicting an exact speed is useful, but an operator needs a decision: “Should I pay attention now?”

So we evaluate not only model ranking, but also how many real congestion onsets are caught early — and how much false-warning burden that creates.

Current development comparison: Caltrans District 12

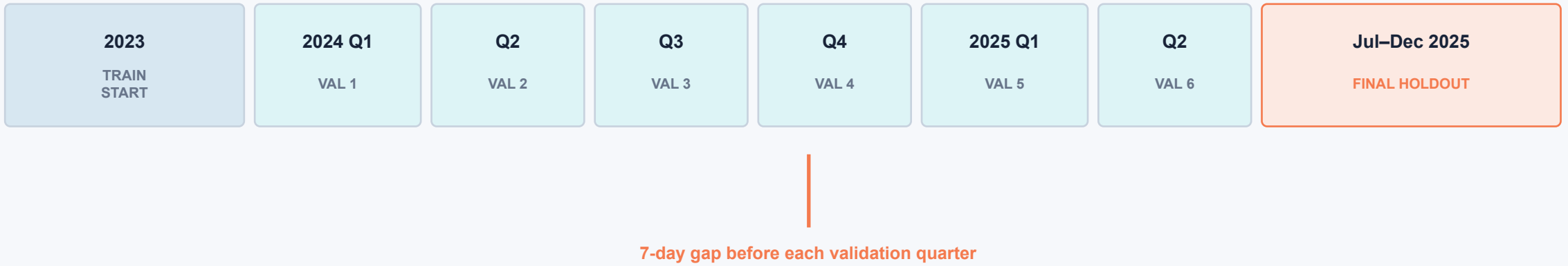
Eligible forecasts are currently uncongested station-times; the system is asked to identify an upcoming onset, not merely confirm an existing jam.

Source: SafeFlow repository README / RESULTS_SUMMARY.md, development snapshot 2026-11-04. Data source: Caltrans PeMS.

Evaluation is designed to look like deployment

Traffic is strongly time-correlated, so a random train/test split would give an unrealistically easy problem.

EXPANDING WALK-FORWARD VALIDATION



What gets tuned?

Model parameters and warning thresholds are chosen using development periods only.

What stays frozen?

Each next-quarter policy is evaluated without retuning on that quarter.

What stays sealed?

July–December 2025 is untouched until the finalist and procedure are frozen.

This is why the current numbers are “development evidence” — not final performance.

Source: SafeFlow README / RESULTS_SUMMARY.md. Six quarterly folds: 2024 Q1–2025 Q2; seven-day training gap; final holdout: Jul–Dec 2025.

Three model families, one fair comparison

All models see the same 29 logical features and the same walk-forward folds; the Transformer additionally sees sequence history.

Random Forest



Many independent trees vote

- Robust tabular baseline
- Captures nonlinear interactions
- Parallel / relatively interpretable

Boosted Trees



Trees correct earlier mistakes

- Strong tabular learner
- Focuses on hard residual patterns
- Current best mean development AP

Transformer



Attention over recent history

- Up to 24 historical feature vectors
- Learns temporal dependencies directly
- Sequence model for evolving traffic

NOVELTY IS THE COMBINATION

A fair model comparison + forward threshold transfer + operational warning-cost analysis

Source: SafeFlow RESULTS_SUMMARY.md. Shared 29-feature representation; Transformer consumes up to 24 historical feature vectors.

Development evidence: ranking quality and warning quality are not the same

The two leading models are close — and the operational tradeoff matters more than a tiny score difference.

DEVELOPMENT ONLY • HOLDOUT UNTOUCHED

1 Ranking quality (mean Average Precision)



AP asks: do higher model scores usually belong to true upcoming congestion?

+0.50 pp

Transformer vs boosted trees

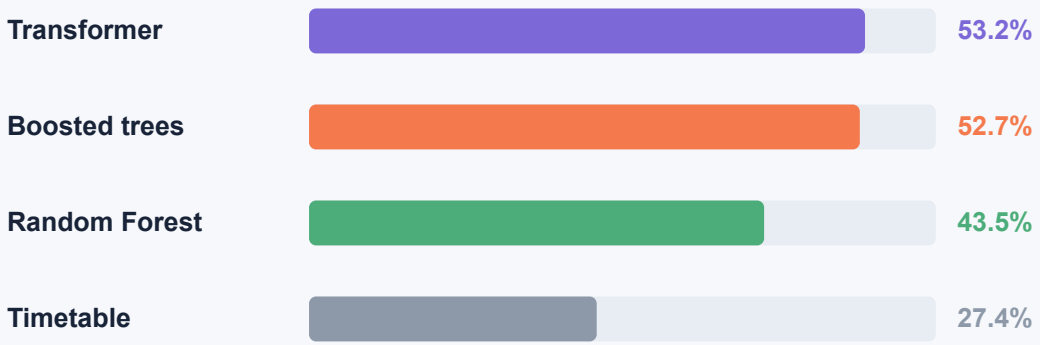
early-catch difference

[-0.66, +1.57]

95% weekly interval

contains zero

2 Early warning at the same intended duration budget



At a 4 wrong-alert-hours/day calibration budget, the Transformer and boosted trees catch about half of onsets at least 15 minutes early.

−2.08

false episodes/day

Transformer − boosted

−0.12 h

wrong-alert hours/day

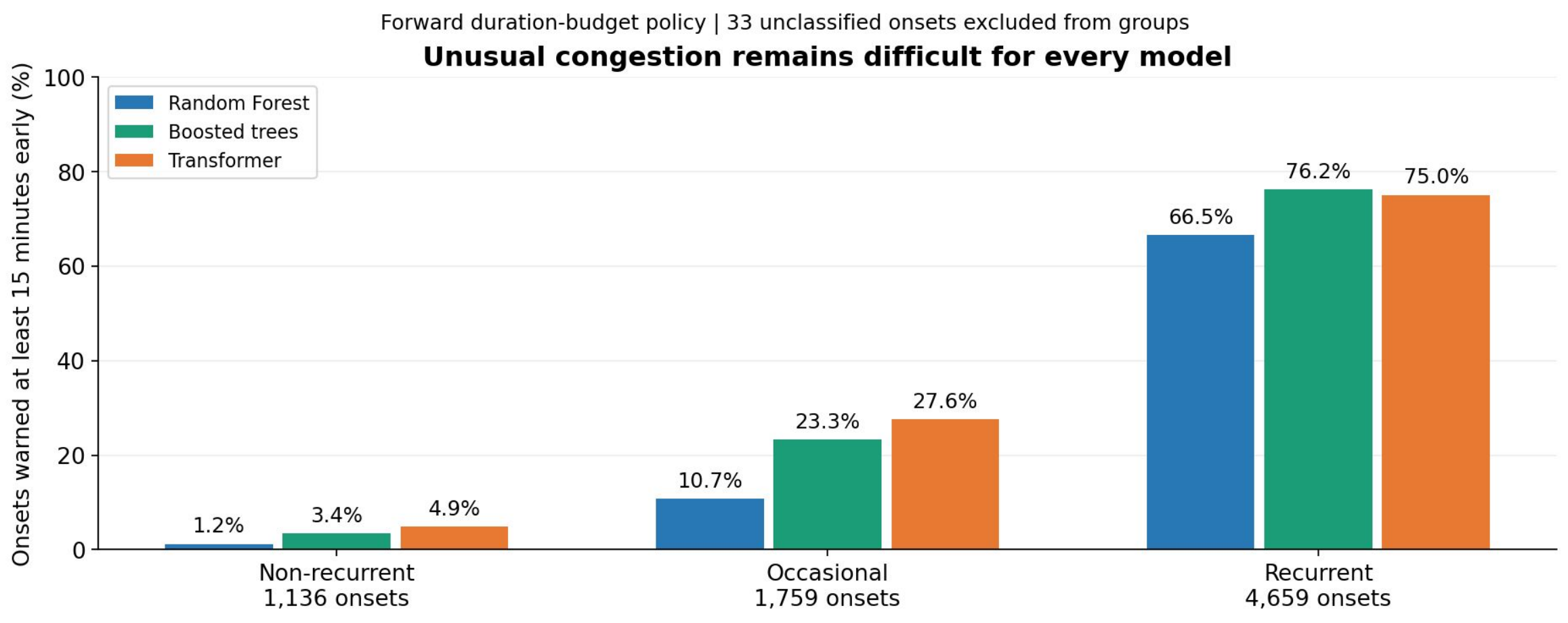
Transformer − boosted

Current decision: keep Transformer and boosted trees as finalists. The evidence does not yet justify declaring a winner.

Source: SafeFlow RESULTS_SUMMARY.md, 2026-11-04. Six-fold AP; five-quarter forward warning-budget validation; paired weekly bootstrap intervals.

The important failure mode: unusual congestion

Aggregate metrics look respectable, but rare non-recurrent onsets remain extremely hard to warn about early.

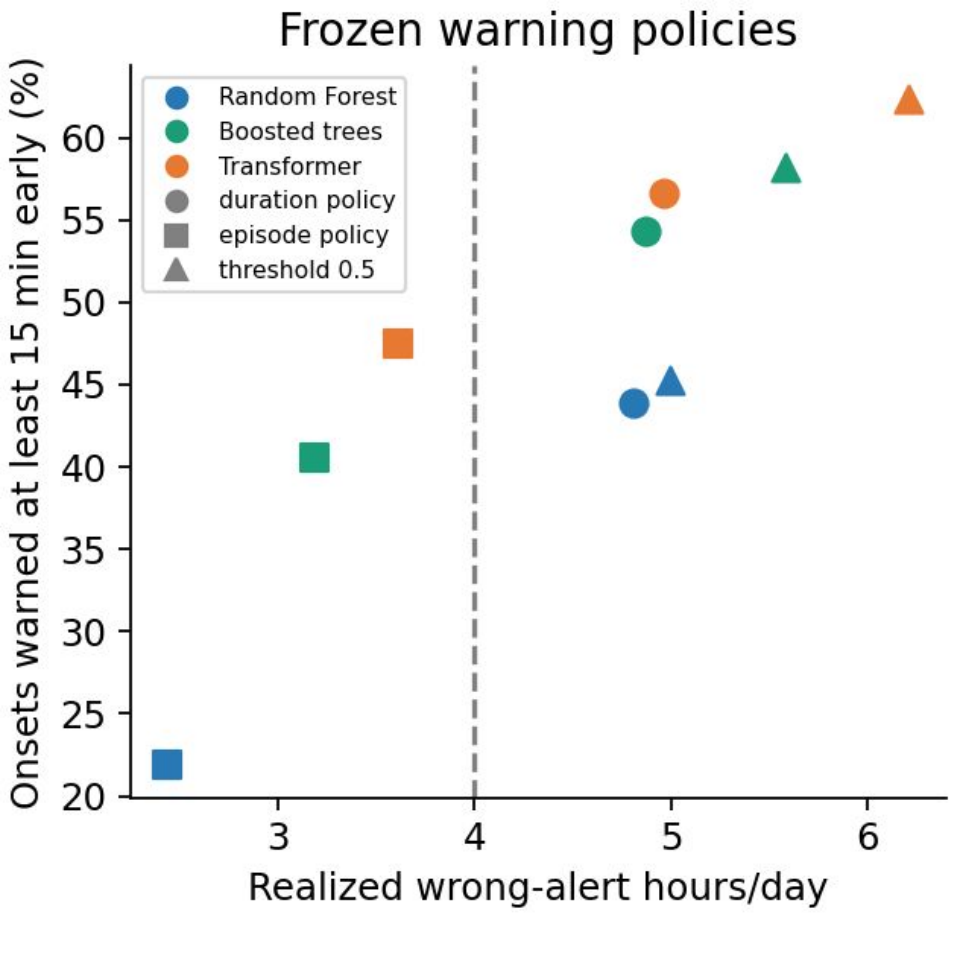


All models miss more than 95% of unusual onsets at a 15-minute lead time.

Source: SafeFlow RESULTS_SUMMARY.md. Forward duration policy; 1,136 non-recurrent onsets across five evaluated quarters.

Final holdout: the tradeoff survives out of sample

Boosted trees rank best; the Transformer catches more onsets early; neither meets the intended warning budget.



Duration policy: higher early catch, but above the 4-hour budget

Transformer: 56.7% early catch, 4.97 h/day
Boosted trees: 54.3% early catch, 4.87 h/day

Circles show the duration policy used for the headline results. Squares and triangles show alternative frozen policies.

No duration-policy model meets the intended budget. Keep both finalists and use a fresh test period for further tuning.

Holdout: Jul–Dec 2025 • 966,000 rows • 125 days • 3,497 onsets • Source: RESULTS_SUMMARY.md / HOLDOUT_REPORT.md

Final Takeaways

Accuracy alone is not enough. A useful congestion-warning system must warn early and keep false alarms manageable.

1 On the unseen holdout (Jul–Dec 2025)

0.741 vs 0.738

Boosted trees rank best

PR-AUC on onsets, vs the Transformer

56.7% vs 54.3%

The Transformer warns earliest

onsets warned ≥15 min ahead, vs boosted trees

≈ 4.9 h/day

Both exceed the 4-hour budget

wrong-alert time, summed over 46 stations

2 Safeflow

1. **Predict the event that matters**
   Congestion within 30 min, from measured readings
2. **Compare models fairly**
   Same features, folds and targets
3. **Test forward in time**
   Thresholds frozen; holdout tested once
4. **Price the cost of warnings**
   Early catch with false episodes and wrong-alert time

Where current models need to improve

- **Unusual jams:** over 90% of non-recurrent onsets are still missed 15 minutes ahead.
- **Stable alert burden:** thresholds set on one season ran about 20% over budget in the next.

Holdout: Jul–Dec 2025 • thresholds frozen before access • Source: HOLDOUT_REPORT.md / RESULTS_SUMMARY.md • Project repository: Erdos-Projects/fall-2026-safeflow